Chase S. Punia

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EDUCATION

Georgia Institute of Technology | B.S. Computer Engineering | Atlanta, GA **Expected Graduation: May 2028**

Threads in Distributed Systems & Software Design | Information Internetworks

WORK EXPERIENCE

Technical Intern II | Adtran *(Software Engineering)* — Norcross, GA ***May – August 2026***

- Built a CLI refactoring tool that automated syntax and keyword modernization across 50,000+ lines of legacy code, cutting a multi-day manual cleanup into a repeatable process that runs in minutes and now serves as the standard tool for other teams
- Re-architected the Jenkins-to-WSL build pipeline across 8 ROADM modules, resolving longstanding deployment failures through systematic root-cause analysis and eliminating a recurring source of build downtime
- Wrote and maintained automated test suites in Robot Framework, validating ROADM module behavior before SIT and catching integration regressions ahead of release

CS Student Assistant | Georgia Tech — Atlanta, GA ***May – June 2026***

- Taught over 30 pre-college students to program autonomous, AI-powered robots across a 3-week Georgia Tech summer camp, designing lesson material that took beginners from zero to a working autonomous navigation program
- Instructed on AI-based computer vision, sensor-based feedback loops, and advanced algorithmic concepts such as Dijkstra's Algorithm and graph traversal

PROJECTS

Free Multi-Agent AI Platform — *Ollama, Qwen3-Coder 32B, Python, Git, Linux, AI Agents*

Designed a hybrid agent architecture that routes planning to frontier APIs and code generation to a locally hosted 32B model, cutting API spend by while preserving quality on multi-step engineering tasks. Implemented task decomposition, agent handoff, and context management across model boundaries, enabling autonomous completion of multi-file refactors without human intervention

PID Controller — *C++, Control Theory, Embedded Systems*

Developed a closed-loop PID controller using integral and differential control to stabilize real-time robotic motion. Tuned feedback loops and motor parameters to cut drivetrain drift, positional error, and overshoot across varying competition environments.

AI Spatial Perception Algorithm — *C++, Fusion 360, IMU/Lidar, Kalman Filtering, Autonomous Navigation*

Fused IMU and lidar data for real-time robot localization within competition fields. Built sensor-driven error-correction and noise-filtering pipelines to improve positional accuracy and autonomous motion planning.

Odometry System & Pure Pursuit Algorithm — *C++, Fusion 360, Kinematics, Path Planning*

Developed odometry algorithms combining encoder data and kinematic modeling to track robot pose and displacement in real time. Designed and 3D-printed the system from scratch, then applied pure pursuit path-tracking to enable smooth, autonomous multi-step navigation.

ACTIVITIES & LEADERSHIP

Competitive Robotics | RoboJackets | RoboDragons | Mentorship | Atlanta, GA 2020 – Current

Competed across VEX, FRC, VEXU, Combat Robotics, and Robo Wrestling at local, regional, and international levels, collectively resulting in **over 60 awards** through engineering and leadership. Experienced in all areas from hardware to software integration.

The Institute of Electrical and Electronics Engineers (IEEE) | Officer **|** Atlanta, GA 2025 – Current

Officer for the world's largest IEEE student branch, managing competition logistics and cross-functional coordination across electrical, mechanical, and software teams. Direct robotics and makerspace operations and co-hosted the IEEE hackathon (robotech2026.netlify.app).

First Robotics Competition (FRC) | GSMST | *Captain* | Lawrenceville, GA 2023 – 2024

Founded GSMST's FRC program from the ground up, and built the team's technical training and recruiting pipeline. Prepared a program to win the state competition with a $6,000 NASA grant which qualified for the world championship in its first competitive season.

HONORS & AWARDS

International Level Awards: Create Award & Innovate Award

Regional Awards & Honors: Tournament Champion, Tournament Finalist, Skills Champion, Design Award & Think Award | Outstanding Computer Science Award | Certificate of Recognition by Gwinnett County (GCPS Board of Education)

SKILLS

Programming Languages: Java, C#, C++, Python, SQL, HTML, Robot Framework

Software Tooling: CLI Tool Development & Scripting, Ollama, Jenkins CI/CD Pipelines, Windows Subsystem for Linux (WSL), Git, Code Review, Debugging, Technical Documentation, Cross-Platform Integration, System Optimization, Gerrit, Polarion & Unity

Algorithms, Controls & Embedded Systems: PID Control, Odometry, Sensor Fusion, Autonomous Navigation, Monte Carlo Localization, Pure Pursuit Algorithms, Embedded Programming, Real-Time Systems, Control Theory, Kinematics & Path Planning, Computer Vision, Sensor-Based Feedback Loops, Motor Control, Sensor Tuning, AI systems & AI Agents deployment